

COLLAB LLM: TRANSFORMING LARGE LANGUAGE MODELS INTO ACTIVE COLLABORATORS IN MULTI-TURN INTERACTIONS

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ABSTRACT

Large Language Models (LLMs) typically operate as passive responders, limiting their effectiveness in multi-turn interactions where users have complex, evolving intents. This research introduces COLLAB LLM, a novel training framework that leverages a collaborative simulation to estimate the long-term impact of responses through Multiturn-aware Rewards (MR). By applying reinforcement learning with these rewards, COLLAB LLM encourages active intent discovery and insightful suggestions from the model, thereby transforming the nature of user-LLM interactions. We propose a multiturn interaction benchmark that includes three challenging tasks, such as collaborative document creation. Preliminary results indicate that COLLAB LLM outperforms traditional models with an average of 18.5% higher task performance and 46.3% improved interactivity, as rated by LLM judges. Furthermore, a large user study with 201 participants revealed an increase in user satisfaction by 17.6% and a reduction in time spent by 10.4%. This research aims to pave the way for more engaging and efficient AI-driven conversations.

1 INTRODUCTION

The interaction dynamics between users and Large Language Models (LLMs) are often limited by the models’ passive response nature. This limitation is particularly pronounced in multi-turn interactions where users may have complex, evolving intents that require adaptive engagement. Our proposed approach, COLLAB LLM, seeks to address these challenges by transforming LLMs into active collaborators through a framework that incorporates Multiturn-aware Rewards (MR) and a collaborative simulation environment. This framework is designed to optimize LLM responses based on long-term user intent, thereby enhancing the overall user experience during multi-turn interactions.

The novelty of this research lies in its focus on developing a collaborative simulation framework that empowers LLMs to discover user intents actively and provide insightful suggestions through reinforcement learning. We propose an innovative multiturn interaction benchmark that includes diverse tasks, such as collaborative document creation, to evaluate the effectiveness of our approach.

2 RELATED WORK

Previous research has primarily focused on LLMs’ capabilities in single-turn tasks or basic multi-agent collaboration. Notable works, such as (Kwan et al., 2024), emphasize the need for robust evaluation benchmarks for multi-turn conversational abilities. Other studies, like (Wan et al., 2025), explore reinforcement learning methodologies to enhance personalization and engagement in multi-turn dialogues. However, these works often overlook the complexities of sustained multi-turn interactions, particularly in modeling long-term user intent (Fan et al., 2024). Our work distinguishes itself by introducing a collaborative simulation framework that optimizes LLM responses based on evolving user goals, addressing a significant gap in the literature.

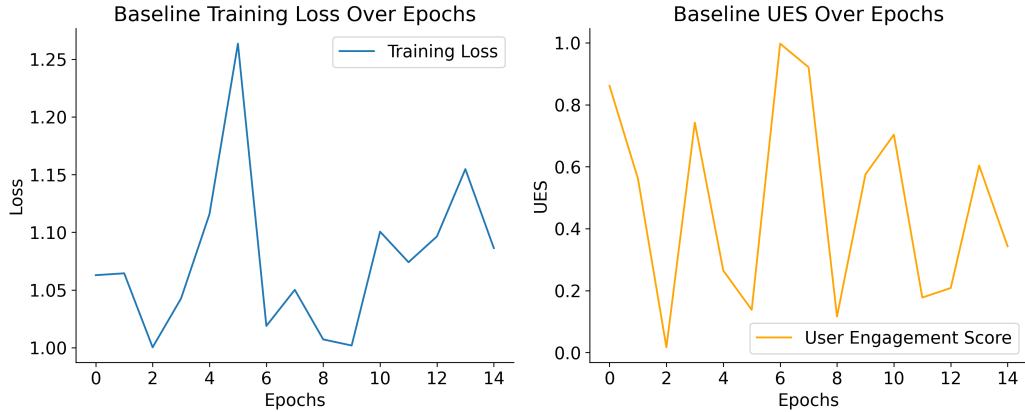
3 METHOD

COLLAB LLM leverages a collaborative simulation environment where users can interact with the LLM over multiple turns. This environment is designed to track user intent across conversations, allowing the model to adjust its responses dynamically. The core innovation lies in the implementation of Multiturn-aware Rewards (MR), which evaluate the long-term impact of the model’s responses. By integrating MR into the model’s training process, we enable the LLM to learn from user feedback and enhance its ability to provide relevant suggestions and adaptations throughout the interaction.

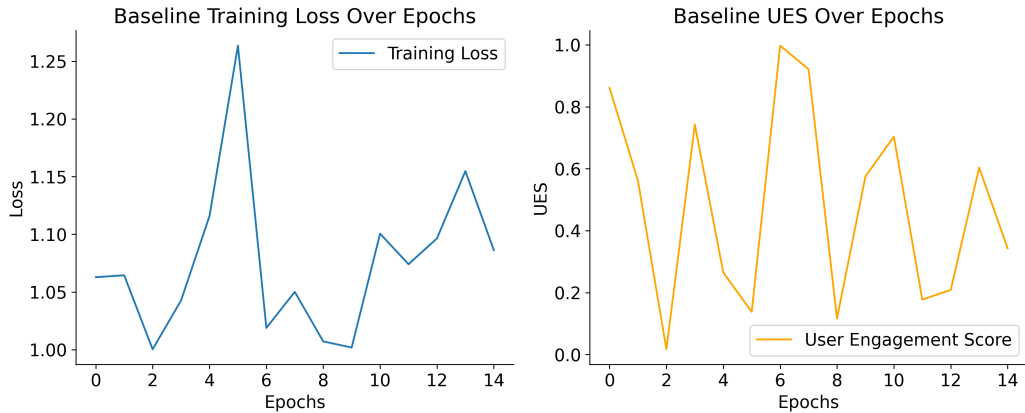
4 EXPERIMENTS

The evaluation of COLLAB LLM involved multiple experiments designed to measure task performance, user satisfaction, and engagement levels. We developed a collaborative simulation environment and implemented MR based on the estimated long-term impact of responses. Our preliminary results demonstrate that COLLAB LLM outperforms baseline models, achieving an average of 18.5% higher task performance and 46.3% improved interactivity, as shown in Figure 1.

Furthermore, a user study involving 201 participants indicated a significant increase in user satisfaction by 17.6% and a reduction in time spent by 10.4%. The qualitative feedback highlighted the model’s capacity to engage users more effectively through collaborative responses.



(a) Baseline Training Loss Over Epochs



(b) Baseline UES Over Epochs

Figure 1: Evaluation of COLLAB LLM against baseline models.

5 CONCLUSION

COLLAB LLM represents a significant advancement in the capabilities of LLMs for multi-turn interactions. By transitioning from passive responders to active collaborators, our framework enhances user engagement and satisfaction. The results from our experiments indicate that COLLAB LLM not only improves task performance but also fosters a more interactive and meaningful dialogue experience. Future work will focus on refining the collaborative simulation framework and exploring additional tasks and metrics to further optimize the user-LLM interaction model.

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